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DRUG DEVELOPMENT

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# An oral allopregnanolone prodrug bypasses liver metabolism via lymphatic transport enabling bioavailability in animals and humans

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**Allopregnanolone, an endogenous neuroactive steroid, has antidepressant and anxiolytic activity but poor oral bioavailability because of substantial first-pass hepatic metabolism. To address the broader challenge of oral bioavailability across clinically validated therapeutics in neuropsychiatry, we developed a lymphatic-targeting prodrug technology platform called Glyph in which a drug of interest is conjugated to a dietary lipid molecule using specifically designed linker chemistry to shift drug absorption to the gut lymphatic system, thus bypassing hepatic metabolism. A series of allopregnanolone prodrugs with different functional linker elements was designed, synthesized, and screened in vitro for gut lymphatic transport and plasma release. Multiple prodrugs achieved therapeutically relevant plasma allopregnanolone concentrations after oral dosing in small- and large-animal models consistent with results from in vitro screening. A triglyceride-mimetic prodrug, GlyphAllo (SPT-300 or Glyph Allopregnanolone), was selected and tested in a phase 1/2a clinical trial (NCT05129865) in 189 healthy participants. Single- and multiple-ascending oral dosing showed that this prodrug could generate therapeutically relevant plasma concentrations in trial participants. GlyphAllo was generally well tolerated and resulted in delivery of allopregnanolone to the systemic circulation and exposure-dependent pharmacodynamic effects via $GABA_A$ (γ-aminobutyric acid type A) receptor positive allosteric modulation. In a subsequent randomized, double-blind, placebo-controlled phase 2a trial using the Trier Social Stress Test (TSST), a validated clinical test for anxiety, GlyphAllo was reported to reduce the salivary cortisol stress response compared with placebo at all post-TSST time points. Collectively, these data support further investigation of GlyphAllo for treating mood and anxiety disorders.**

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## INTRODUCTION

Many promising therapeutics have been abandoned because of physiochemical or biological factors that limit oral bioavailability such as poor absorption or extensive first-pass hepatic metabolism (*1*–*3*). Drug delivery to the systemic circulation via the lymphatic system avoids first-pass hepatic metabolism and provides several advantages over conventional oral administration, including increased oral bioavailability and enhanced delivery to lymph nodes and immune cells (*1*, *4*–*6*). However, few small-molecule drugs are sufficiently lipophilic to incorporate into endogenous lymph lipid transport pathways (*4*). We have recently advanced a platform called Glyph that uses a biomimetic lipid prodrug approach to direct small-molecule drugs into the intestinal lymphatic system, bypassing first-pass hepatic metabolism and enhancing oral bioavailability (*7*, *8*).

In conventional oral drug transport, small molecules are absorbed from the intestinal lumen into gut enterocytes and then trafficked into the portal blood and exposed to first-pass metabolism before reaching the systemic circulation (*4*). In contrast, ingested dietary triglycerides (TGs) are hydrolyzed by pancreatic lipases in the small intestine into monoglyceride (MG) and fatty acid (FA) intermediates that are absorbed into gut enterocytes, reesterified to TGs, incorporated into lipoproteins, and trafficked through the lymphatic system to the systemic circulation (fig. S1) (*9*).

Although small-molecule drugs typically do not have the physicochemical properties required for efficient lymphatic transport (*4*), lipid-mimetic prodrugs can use the same absorption and trafficking pathways as dietary TGs (*1*, *5*, *6*). The possibility of using TG-drug conjugates to promote lymphatic absorption has been explored since the late 1970s (*10*–*13*). These efforts demonstrated enhanced bioavailability in rodents (*10*, *11*, *14*–*16*) but were not translated into enhanced bioavailability in humans (*1*, *12*, *17*).

We have developed a rational design approach to increase the lymphatic transport of oral TG-mimetic prodrugs, enhance bioavailability, and enable systemic drug release through (i) the addition of an optimized linker scaffold connecting the drug and glycerol moieties to promote gastrointestinal (GI) stability and lymph-directed enterocyte processing (*18*–*23*) and (ii) the use of a self-immolative (SI) group to facilitate release of the active pharmaceutical ingredient into the systemic circulation (*19*–*21*, *24*). This design approach has shown promise in rodents and dogs and provides rationale for translation to higher-order animal species (*19*, *25*).

Here, we have translated this approach into humans to enable higher oral bioavailability with a well-characterized and therapeutically efficacious compound that has been previously limited by extensive first-pass hepatic metabolism (*26*–*28*). This compound was the endogenous neurosteroid allopregnanolone (ALLO), a potent positive allosteric modulator of γ-aminobutyric acid (GABA) type A ($GABA_A$) receptors (*28*, *29*). $GABA_A$ receptors and ALLO have been implicated in the underlying pathophysiology of neurologic and neuropsychiatric conditions including depression, anxiety, epileptic disorders, essential tremor and sleep disorders, and fragile

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X-associated tremor/ataxia syndrome (*27*, *30*–*32*). Studies in rodents have shown the anxiolytic and antistress behavioral effects of ALLO (*26*, *27*) and other $GABA_A$ receptor positive allosteric modulators (*28*, *33*). An intravenous infusion of ALLO (brexanolone) and an oral synthetic neuroactive steroid (zuranolone) have received regulatory approval in the United States to treat postpartum depression, and an oral synthetic neuroactive steroid (ganaxolone) has received regulatory approval for selected seizure disorders (*34*–*38*). However, brexanolone requires a continuous 60-hour intravenous infusion because of low oral bioavailability and is not commercially available at present, whereas oral synthetic neuroactive steroids can be limited by pharmacologic properties that are not identical to those of ALLO (*28*, *36*, *39*–*41*). Given the therapeutic potential of ALLO across psychiatric and neurologic disorders (*28*), an oral ALLO formulation is desirable, but its poor oral bioavailability due to extensive first-pass hepatic metabolism has made oral ALLO delivery challenging (*28*).

Herein, we report the synthesis, evaluation, optimization, and initial clinical development of a TG-mimetic prodrug of ALLO using the Glyph platform. ALLO prodrugs with different linker chemistries demonstrated lymphatic transport in rodents, differential in vitro plasma release, and translation from initial rodent studies into higher-order species. Of these optimized orally bioavailable ALLO prodrugs, GlyphAllo (SPT-300 or Glyph Allopregnanolone; formerly LYT-300) was selected for advancement into a phase 1/2a clinical trial in healthy human volunteers.

## RESULTS

### Prodrug design, evaluation, and clinical candidate selection

#### *Prodrug design and synthesis*

ALLO lipid prodrugs were designed such that ALLO was attached to a variety of lymph-directing glyceride moieties via an assortment of diacid linkers (Fig. 1A). The linker was designed to be modifiable in

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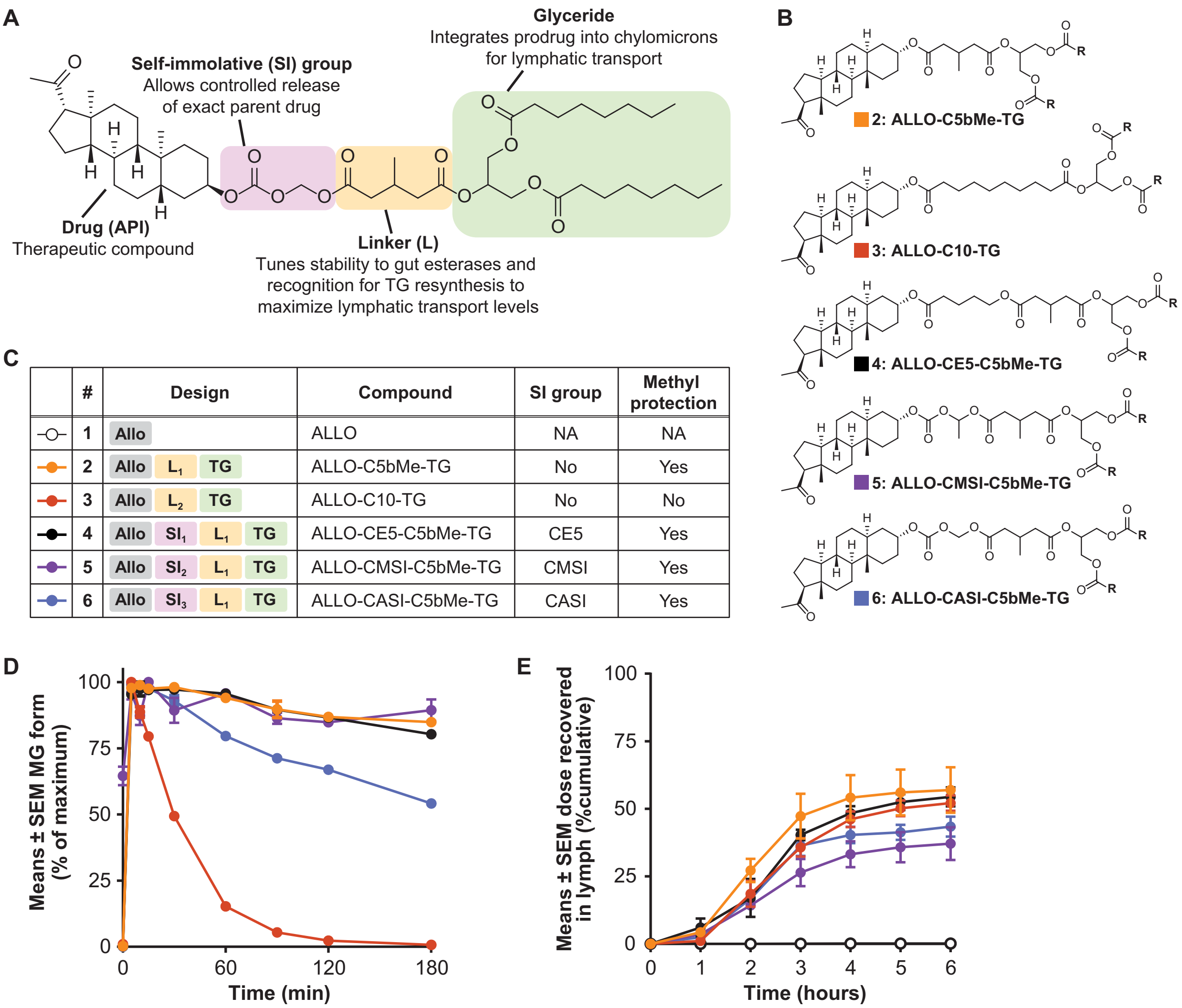


| | # | Design | Compound | SI group | Methyl protection |
|---|---|---|---|---|---|
| –○– | 1 | Allo | ALLO | NA | NA |
| –●– (orange) | 2 | Allo L$_1$ TG | ALLO-C5bMe-TG | No | Yes |
| –●– (red) | 3 | Allo L$_2$ TG | ALLO-C10-TG | No | No |
| –●– (black) | 4 | Allo SI$_1$ L$_1$ TG | ALLO-CE5-C5bMe-TG | CE5 | Yes |
| –●– (purple) | 5 | Allo SI$_2$ L$_1$ TG | ALLO-CMSI-C5bMe-TG | CMSI | Yes |
| –●– (blue) | 6 | Allo SI$_3$ L$_1$ TG | ALLO-CASI-C5bMe-TG | CASI | Yes |

**Fig. 1. Design and synthesis of ALLO-TG prodrugs.** (**A**) Design of the ALLO lipid prodrug. (**B**) Structures and (**C**) component summary of the five prodrugs selected for evaluation. Fatty acid chains in prodrugs (indicated by "R" in each chemical structure) varied (see table S1). (**D**) In vitro prodrug stability after pancreatin digestion in simulated intestinal fluid was assessed (means ± SEM, $n = 3$) (*19*). (**E**) After prodrug infusion directly into the duodenum of the anesthetized rat, gut-draining lymph was collected from the cannula, and cumulative mass of prodrug-related compounds was determined to calculate the proportion of the total administered dose recovered in the lymph (means ± SEM, $n = 3$). API, active pharmaceutical ingredient; CE, cyclizing ester; Me, methyl; NA, not applicable.

terms of its carbon chain length and substitution pattern to modulate function and allow for incorporation of an SI group to facilitate active pharmaceutical ingredient release into the systemic circulation (*19*). In the series synthesized, five ALLO prodrugs are described (Fig. 1, B and C), illustrating the impact of both substituted and unsubstituted linkers and a range of SI groups (figs. S2 and S3). Prodrugs ALLO-C5bMe-TG (**2**) (*42*) and ALLO-C10-TG (**3**) were selected as structures with and without methyl protection, respectively, and with linkers that lacked an SI group, whereas the linkers of prodrugs ALLO-CE5-C5bMe-TG (**4**), ALLO-CMSI-C5bMe-TG (**5**), and ALLO-CASI-C5bMe-TG (**6**) were all methyl protected and contained different SI groups. Because the C10 linker maintained an overall length similar to those of the SI-C5bMe combinations, prodrug **3** was also selected as a comparator to the SI-containing prodrugs **4**, **5**, and **6**.

### *Prodrug stability in simulated intestinal fluid*

TG-based prodrugs are processed similarly to dietary TGs, with FAs hydrolyzed from the TG backbone before intestinal absorption to reveal an MG-like prodrug derivative (*43*). The stability of this MG-like derivative in the gut lumen is critical to allow intact absorption into the gut enterocytes and entry into lymphatic transport pathways (*17*). Prodrug stability was screened in vitro using simulated intestinal fluid containing pancreatin as a surrogate for the enzymatic contents of the intestine. After a 3-hour incubation with simulated intestinal fluid, ALLO-MG derivatives generated from prodrugs with a C5bMe linker were well protected, with >60% remaining at 120 min (Fig. 1D). In contrast, the MG **3a** generated from prodrug **3** (fig. S2) was hydrolyzed rapidly, with <20% remaining after 60 min. Among the prodrugs containing SI C5bMe linkers, the MG **6a** derivative containing the carboxyacetal self-immolative (CASI) group was comparatively the most labile, although the 60% that remained after 3 hours was sufficient to allow enterocyte absorption of the ALLO-MG derivative hydrolysis product and to promote lymphatic transport (*44*, *45*).

### *Prodrug transport into mesenteric lymph in anesthetized rats*

Once the MG and FA components of dietary TGs are absorbed into enterocytes, they are reesterified into TGs, packaged into chylomicrons, and secreted into mesenteric lymph (fig. S1, B and C). Next, the process of prodrug transport into the lymphatic system was assessed in anesthetized rats in which mesenteric lymph was cannulated (*46*). In this experiment, 40 to 60% of the administered dose of each of the five prodrugs was recovered in mesenteric lymph fluid as the reesterified glyceride (Fig. 1E), with negligible differences among prodrugs. From 4 hours onward, very little (<6%) additional prodrug was recovered in mesenteric lymph fluid (Fig. 1E). Despite poor luminal stability of MG **3a** (*42*) (Fig. 1D), the unsubstituted C10 linker did not appear to impede lymphatic transport of prodrug **3**, and the proportion of the recovered dose in the lymph was comparable to that seen for compounds with C5bMe linkers (Fig. 1E). This may be due to enzyme dilution during duodenal infusion that may not be representative of that encountered after oral administration. ALLO was undetectable in lymph samples after administration of unmodified active pharmaceutical ingredient, consistent with poor transport into the lymph.

### *In vitro plasma release and in vivo plasma pharmacokinetics of prodrugs in animal models*

After secretion into the lymph, chylomicrons move through mesenteric lymph ducts and the thoracic duct to enter the systemic circulation near the confluence of the jugular and subclavian veins (fig. S1D). In the systemic circulation, chylomicron TGs are hydrolyzed by lipoprotein lipase to liberate FAs and MGs (*47*, *48*). Prodrug conversion to active pharmaceutical ingredient in plasma was assessed across a range of animal species. Prodrugs were incubated in plasma with lipoprotein lipase to provide the necessary initial cleavage of the FA chains, forming the first key intermediate ALLO-MG, which was then hydrolyzed by plasma esterases at either end of the linker to release a second key intermediate, ALLO-linker acid (ALLO-LA), or the ALLO active pharmaceutical ingredient (fig. S1D).

Incubation of each of the five prodrugs with lipoprotein lipase in rat plasma showed rapid conversion of all prodrugs into their MG forms (Fig. 2A). Of these, MGs **3a** through **6a** were rapidly hydrolyzed, with <50% remaining after 60 min. In contrast, the directly attached C5bMe MG **2a** underwent slow conversion into the ALLO-LA intermediate **2b** (Fig. 2B) but did not produce a detectable amount of ALLO (Fig. 2C), suggesting esterase cleavage at the TG end but not the ALLO end of the linker. Similar hydrolysis profiles were seen for the directly attached C10 and CE5-C5bMe MGs **3a** and **4a**, respectively, with substantial amounts of each ALLO-LA intermediate detected (Fig. 2B) but, in this case, with some additional ALLO generation (≈25% at 180 min) (Fig. 2C). This finding suggests that MGs **3a** and **4a** are likely being cleaved more rapidly at the TG end of the linker but that both the ALLO-MG and ALLO-LA intermediates of each prodrug can be cleaved at the active pharmaceutical ingredient end to release ALLO. In the third hydrolysis profile observed, the MGs **5a** and **6a** derived from the carboxy(methylacetal) self-immolative (CMSI) and CASI prodrugs were rapidly converted to ALLO (≈75% in 30 min) (Fig. 2C), with no detectable ALLO-LA intermediate produced from either prodrug (Fig. 2B). After oral dosing in rats, no measurable plasma exposure was seen in vivo with either ALLO or the directly attached prodrug **2** (Fig. 2D), indicating low lymphatic transport or poor in vivo release. In contrast, substantial systemic exposure of ALLO was evident after administration of the other four prodrugs, reflecting robust lymphatic transport and effective in vivo release.

In vivo distribution of ALLO in ALLO-CASI-C5bMe-TG-octanoate (prodrug **6**) was measured in rats using quantitative whole-body autoradiography after the dosing of prodrug **6** radiolabeled either on ALLO or the linker moiety (fig. S4). When radiolabeled on ALLO, radioactivity was associated with the lymph duct between 1 and 24 hours postdose and with white and brown fat (fig. S4A). By 72 hours postdose, radioactivity was measurable only in the GI tract, and by 168 hours after dosing, radioactivity was below the lower limit of quantification or not detected for all tissues analyzed. Elimination was primarily via the feces, consistent with ALLO. When radiolabeled on the linker, radioactivity was associated with the lymph duct and spleen between 1 and 2 hours postdose (fig. S4B). By 72 hours, radioactivity was only measurable in white fat, and by 168 hours after dosing, radioactivity was below the lower limit of quantification or not detected for all tissues analyzed. Elimination was primarily renal, consistent with a relatively polar small molecule.

### *In vitro plasma release and in vivo plasma pharmacokinetics of prodrugs in large-animal models*

Translation of these findings from rats to higher-order animals was conducted in dogs (Fig. 2, E to H) and nonhuman primates (NHPs) (Fig. 2, I to L). Given its lack of in vivo exposure in rats, the directly attached prodrug **2** was not tested. For the other four prodrugs, in vitro and in vivo profiles were similar between dogs and NHPs. Hydrolysis of the ALLO-MG intermediate was markedly slower for

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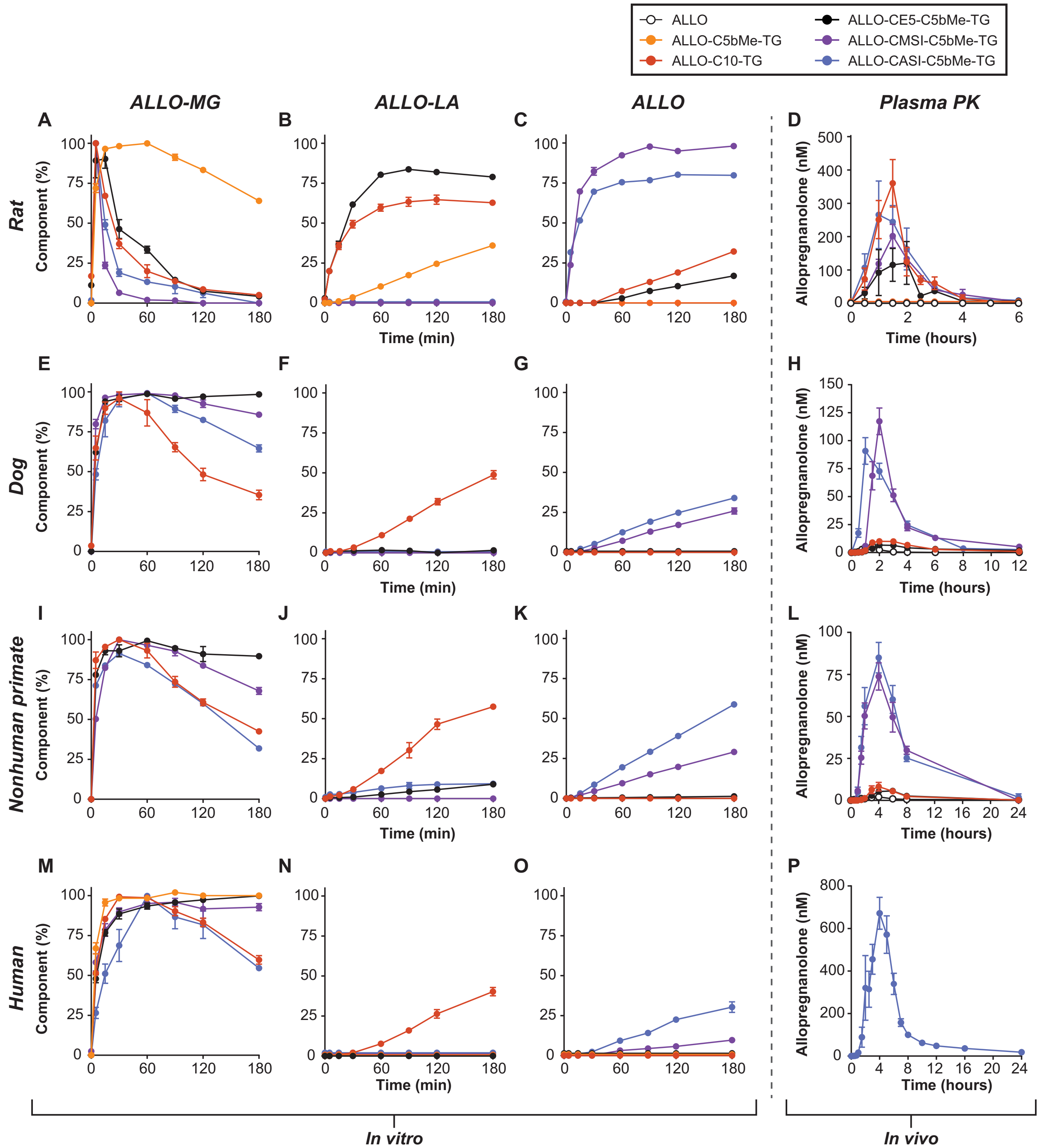


**Fig. 2. Prodrug conversion and plasma PK across four animal species.** Prodrugs were incubated in vitro with lipoprotein lipase in rat (**A** to **C**), dog (**E** to **G**), NHP (**I** to **K**), and human (**M** to **O**) plasma, and relative concentrations (means ± SEM, $n = 3$) of two key intermediates, ALLO-MG [(A), (E), (I), and (M)] and ALLO-LA [(B), (F), (J), and (N)], and ALLO [(C), (G), (K), and (O)] were assessed. ALLO data are presented as an absolute percentage of starting prodrug concentration based on quantification against a standard curve. ALLO-MG data are presented as a normalized percentage (100% defined as the largest LC-MS peak area in a given experiment assuming rapid quantitative conversion to MG form); ALLO-LA data are presented as a normalized percentage of starting prodrug concentration by subtraction of the ALLO and ALLO-MG percentages at corresponding time points. ALLO plasma exposure (means ± SEM) after oral gavage of prodrug was measured in rats (**D**) (**1** and **3**, $n = 4$; **2**, $n = 3$; **4**, $n = 5$; **5**, $n = 6$; **6**, $n = 3$), dogs (**H**) (**1**, **3**, **4**, and **5**, $n = 4$; **6**, $n = 10$), NHPs (**L**) (**1**, **3**, and **4**, $n = 6$; **5** and **6**, $n = 5$), and humans (**P**) (**6**, $n = 9$), with concentrations dose-normalized to an equivalent ALLO dose (2 mg/kg). Because no in vivo exposure for prodrug **2** was seen in rats, prodrug **2** was not tested in dogs or NHPs. For visualization, the following datasets have been nudged upward on their respective $y$ axes by <2 units: **1** [orange; (D), (N), and (O)], **4** [black; (G) and (O)], **5** [purple; (N)], and **6** [blue; (B), (F), and (N)]; for all nudged datasets, $y = 0$ at all time points. Data were collated from separate experiments in which fatty acid chains in prodrugs varied (table S1). LA, linker acid; LC-MS, liquid chromatography–mass spectrometry.

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prodrugs **4**, **5**, and **6** in plasma from higher-order species (Fig. 2, E and I) compared with rat plasma (Fig. 2A), which was expected given observations of higher plasma esterase activity in rodents versus higher-order species (*49*). MG **4a** displayed the greatest stability and showed little conversion to ALLO-LA **4b** or ALLO in either dog or NHP plasma, whereas the other ALLO-MG intermediates showed some level of drug release. The directly attached MG **3a** was hydrolyzed relatively quickly (<50% remaining after 180 min) and converted into ALLO-LA **3b** but not ALLO in both dog and NHP plasma. In contrast, hydrolysis of the SI-containing MGs **5a** and **6a** was accompanied by ALLO formation, with only a small amount of ALLO-LA **6b** observed in NHPs. This suggested that the CMSI- and CASI-containing ALLO prodrugs **5** and **6** had the desired in vitro drug release profile in which the MG intermediates **5a** and **6a** underwent preferential cleavage of the linker at the active pharmaceutical ingredient ends, resulting in the desired release of ALLO, consistent with data obtained in rats.

In both dogs and NHPs, in vivo systemic exposure after oral dosing (Fig. 2, H and L) closely followed the patterns of in vitro ALLO release (Fig. 2, G and K), with robust oral exposure seen for prodrugs **5** and **6** but not prodrugs **3** or **4**. Orally dosed ALLO showed almost no exposure in either animal species. ALLO was also dosed intravenously in both dogs and NHPs to enable calculation of apparent oral bioavailability, which was ≈25 to 35% for both prodrugs **5** and **6** and <5% for both prodrugs **3** and **4**.

### *In vitro plasma release and in vivo plasma pharmacokinetics of prodrugs in humans*

Lipid prodrugs **3**, **4**, **5**, and **6** were assessed in a human plasma incubation study. Results closely mimicked the data for dogs and NHPs (Fig. 2, M to O), consistent with the similarity in plasma esterase activity (*49*). ALLO release was the greatest with prodrug **6** (≈25% in 3 hours), lower with prodrug **5** (≈10% in 3 hours), and undetectable for prodrugs **3** and **4** (Fig. 2O). As observed in dogs and NHPs, hydrolysis of the directly attached MG **3a** generated the ALLO-LA intermediate **3b** but little ALLO (Fig. 2N). Consistent with the plasma release data, robust systemic exposure was observed for prodrug **6** after oral dosing in humans (Fig. 2P).

These in vitro and in vivo data are summarized and contextualized in Fig. 3. Of the five prodrugs evaluated, all underwent lymphatic transport, and four (prodrugs **3**, **4**, **5**, and **6**) showed conversion and ALLO release in rats; the highest bioavailability of ALLO after oral administration in higher-order animals was seen with prodrugs **5** and **6**. During prodrug development, the FA moiety on each prodrug was varied among palmitate, oleate, or octanoate versions (fig. S2 and table S1); however, as shown in fig. S5, these three FA variants had similar downstream properties. On the basis of these data, prodrug **6** was selected as the GlyphAllo clinical candidate and advanced into a multipart, first-in-human clinical trial to investigate the effects of oral GlyphAllo on safety, tolerability, pharmacokinetics (PK) (single-ascending dose, multiple-ascending dose, and food effect), pharmacodynamics (PD), and cortisol response in healthy volunteers subjected to acute social stress.

## Phase 1/2a clinical trial of GlyphAllo in healthy volunteers

### *Clinical trial participant characteristics*

In this phase 1/2a trial, 189 healthy volunteers were enrolled across part 1 (single-ascending dose; $n = 30$), part 2 (food effect; $n = 9$), part 3 (multiple-ascending dose; $n = 60$), and part 4 [Trier Social Stress Test (TSST); $n = 90$] (fig. S6). A total of 10 healthy volunteers withdrew from the trial (fig. S6): 4 due to participant withdrawal of consent [cohort 1, $n = 2$ (placebo, no reason provided; 140 mg of GlyphAllo, due to undisclosed personal reasons); cohort 4, $n = 1$ (250 mg of GlyphAllo, due to bereavement); cohort 8, $n = 1$ (375 mg of GlyphAllo, due to clinical/environmental factors)], 2 due to being COVID-19 close contacts [cohort 1, $n = 1$ (placebo); cohort 2b, $n = 1$ (750 mg of GlyphAllo)], 2 due to treatment-emergent adverse events (TEAEs) from COVID-19 after 3 days [cohort 2b, $n = 1$ (placebo)] and 5 days [cohort 4, $n = 1$ (250 mg of GlyphAllo)] of dosing, 1 due to physician decision [cohort 3, $n = 1$ (140 mg of GlyphAllo, due to inability to be cannulated; judged by an investigator as unrelated to treatment)], and 1 due to a hand fracture (judged by an investigator as unrelated to treatment) and concomitant medications that violated predosing requirements [cohort 2b, $n = 1$ (750 mg of GlyphAllo)]. Demographic and baseline characteristics were generally comparable across all cohorts (Table 1), median age ranged from 25 to 31 years, and most healthy volunteers were male and white.

### *Safety and tolerability*

Of 189 healthy volunteers enrolled across all four parts of the trial, 127 (67.2%) reported 493 TEAEs (Table 2). One grade 3 serious TEAE (hand fracture that occurred postdischarge), determined by

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| # | Design | Compound | SI group | Methyl protection |
|---|---|---|---|---|
| 1 | Allo | ALLO | NA | NA |
| 2 | Allo $L_1$ TG | ALLO-C5bMe-TG | No | Yes |
| 3 | Allo $L_2$ TG | ALLO-C10-TG | No | No |
| 4 | Allo $SI_1$ $L_1$ TG | ALLO-CE5-C5bMe-TG | CE5 | Yes |
| 5 | Allo $SI_2$ $L_1$ TG | ALLO-CMSI-C5bMe-TG | CMSI | Yes |
| 6 | Allo $SI_3$ $L_1$ TG | ALLO-CASI-C5bMe-TG | CASI | Yes |

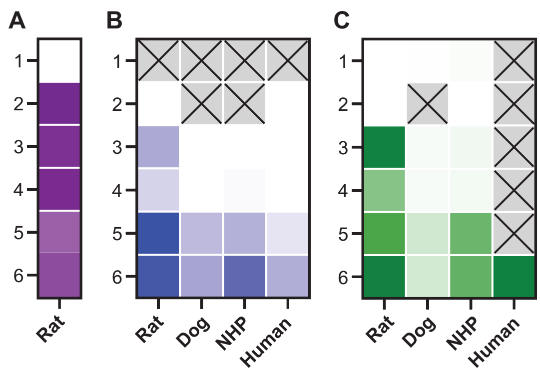


**Fig. 3. Summary of prodrug conversion and plasma PK across four species.** (**A**) Total percentage of administered dose collected in rat lymphatics 6 hours after dosing [minimum value (white), 0.15%; maximum value (dark purple), 56.97%]. (**B**) In vitro conversion of the prodrug to ALLO in the plasma of various species at 3 hours and 37°C [minimum value (white), 0%; maximum value (dark blue), 99%]. (**C**) Dose-normalized plasma exposure (AUC/D) of ALLO [the minimum value (white) is below the limit of quantification; maximum value (dark green), 1579.5 nM per hour per kg/mg]. Cells in gray indicate that the compound was not assayed in the species indicated. Plasma exposure was normalized for body surface area using conventional allometric scaling factors (human, 1.0; dog, 0.541; NHP, 0.324; rat, 0.162). AUC, area under the concentration-time curve; CA, carboxyacetal; CM, carboxy(methylacetal); D, dose; L, linker.

**Table 1. Demographic characteristics at baseline for phase 1/2a trial in healthy volunteers (safety population).** BMI, body mass index; FE, food effect; HV, healthy volunteer; MAD, multiple-ascending dose; PBO, placebo; SAD, single-ascending dose.

| Characteristic | Part 1: SAD | | | Part 2: FE | | Part 3: MAD | | | | | Part 4: TSST | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cohort 1 (PBO, 70, 140, and 280 mg) *n* = 11 | Cohort 2 (PBO, 280, 420, and 560 mg) *n* = 9 | Cohort 2b (PBO, 750 and 1000 mg) *n* = 10 | Cohort 3 (140 mg) *n* = 9 | Pooled PBO *n* = 15 | Cohort 4 (250 mg QAM) *n* = 9 | Cohort 5 (375 mg QAM) *n* = 9 | Cohort 6 (500 mg QAM) *n* = 9 | Cohort 7 (375 mg QHS *n* = 9 | Cohort 8 (375 mg QHS) *n* = 9 | Cohort 9a (PBO) *n* = 10 | Cohort 9b (PBO) *n* = 39 | Cohort 9b (375 mg) *n* = 41 |
| *Age, years* | | | | | | | | | | | | | |
| Mean (SD) | 28.6 (8.4) | 32.0 (8.5) | 27.7 (8.2) | 28.3 (7.0) | 28.6 (8.8) | 30.0 (5.8) | 32.1 (8.4) | 30.2 (9.2) | 30.2 (10.7) | 28.9 (12.4) | 29.6 (6.6) | 30.2 (9.6) | 29.7 (10.4) |
| Median (range) | 25.0 (21, 48) | 30.0 (21, 48) | 25.5 (19, 46) | 28.0 (18, 40) | 26.0 (18, 47) | 28.0 (21, 39) | 31.0 (24, 50) | 29.0 (19, 49) | 29.0 (19, 54) | 23.0 (19, 53) | 29.5 (21, 39) | 26.0 (19, 52) | 25.0 (19, 54) |
| *Sex,* n *(%)* | | | | | | | | | | | | | |
| Female* | 4 (36.4) | 2 (22.2)* | 1 (10.0) | 2 (22.2) | 5 (33.3) | 2 (22.2) | 5 (55.6) | 4 (44.4) | 3 (33.3) | 5 (55.6)* | 4 (40.0) | 8 (20.5) | 10 (24.4) |
| Male | 7 (63.6) | 7 (77.8) | 9 (90.0) | 7 (77.8) | 10 (66.7) | 7 (77.8) | 4 (44.4) | 5 (55.6) | 6 (66.7) | 4 (44.4) | 6 (60.0) | 31 (79.5) | 31 (75.6) |
| *Ethnicity,* n *(%)* | | | | | | | | | | | | | |
| Hispanic/Latino | 3 (27.3) | 0 | 1 (10.0) | 2 (22.2) | 2 (13.3) | 1 (11.1) | 1 (11.1) | 0 | 2 (22.2) | 0 | 1 (10.0) | 6 (15.4) | 3 (7.3) |
| Not Hispanic/ Latino | 8 (72.7) | 9 (100) | 9 (90.0) | 7 (77.8) | 12 (80.0) | 8 (88.9) | 5 (55.6) | 8 (88.9) | 7 (77.8) | 9 (100) | 8 (80.0) | 32 (82.1) | 38 (92.7) |
| Other/unknown/ not reported | 0 | 0 | 0 | 0 | 1 (6.7) | 0 | 3 (33.3) | 1 (11.1) | 0 | 0 | 1 (10.0) | 1 (2.6) | 0 |
| *Race,* n *(%)* | | | | | | | | | | | | | |
| American Indian/ Alaskan Native | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 (5.1) | 1 (2.4) |
| Asian | 3 (27.3) | 1 (11.1) | 3 (30.0) | 1 (11.1) | 2 (13.3) | 3 (33.3) | 1 (11.1) | 3 (33.3) | 2 (22.2) | 0 | 1 (10.0) | 6 (15.4) | 10 (24.4) |
| Black or African American | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 (2.4) |
| Native Hawaiian/ Pacific Islander | 0 | 0 | 0 | 0 | 0 | 1 (11.1) | 0 | 1 (11.1) | 0 | 0 | 0 | 0 | 0 |
| White | 6 (54.5) | 8 (88.9) | 7 (70.0) | 7 (77.8) | 12 (80.0) | 4 (44.4) | 5 (55.6) | 4 (44.4) | 4 (44.4) | 7 (77.8) | 9 (90.0) | 30 (76.9) | 29 (70.7) |
| Other/multiple† | 2 (18.2) | 0 | 0 | 1 (11.1) | 1 (6.7) | 1 (11.1) | 3 (33.3) | 1 (11.1) | 3 (33.3) | 2 (22.2) | 0 | 1 (2.6) | 0 |
| *Body mass index, kg/m²* | | | | | | | | | | | | | |
| BMI, mean (SD) | 24.8 (2.1) | 23.6 (3.7) | 25.0 (2.2) | 26.1 (3.1) | 23.7 (3.0) | 23.8 (3.9) | 23.9 (2.8) | 24.6 (3.2) | 24.7 (3.8) | 24.6 (3.0) | 25.8 (3.1) | 24.5 (2.7) | 24.8 (2.4) |

*All females were of childbearing potential except for one healthy volunteer in cohorts 2 and 8. †"Other" includes those who identified as Aboriginal, Brazilian, Latin American, Mexican, Middle Eastern, or native Brazilian. "Multiple" includes those who identified as Asian and white, American Indian/Alaskan Native and white, or Black/African American and white.

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**Table 2. Summary of safety outcomes in the phase 1/2a trial in healthy volunteers (safety population).** The safety population was defined as all healthy volunteers who received any amount of the study drug (GlyphAllo or placebo). A total of 122 distinct participants received ≥1 dose of GlyphAllo in the phase 1/2a trial across parts 1 to 4. However, a total of 139 participants appear in the safety data tables under GlyphAllo because of their inclusion in ≥1 cohort per the crossover design. FE, food effect; HV, healthy volunteer; *m*, number of events; *n*, number of HVs.

| ***n* (%); *m*** | **Part 1: SAD** | | | **Part 2: FE** | | **Part 3: MAD** | | | | | **Part 4: TSST** | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Cohort 1 (PBO, 70, 140, and 280 mg) *n* = 11** | **Cohort 2 (PBO, 280, 420, and 560 mg) *n* = 9** | **Cohort 2b (PBO, 750 and 1000 mg) *n* = 10** | **Cohort 3 (140 mg) *n* = 9** | **Pooled PBO *n* = 15** | **Cohort 4 (250 mg QAM) *n* = 9** | **Cohort 5 (375 mg QAM) *n* = 9** | **Cohort 6 (500 mg QAM) *n* = 9** | **Cohort 7 (375 mg QHS *n* = 9** | **Cohort 8 (375 mg QHS) *n* = 9** | **Cohort 9a (PBO) *n* = 10** | **Cohort 9b (PBO) *n* = 39** | **Cohort 9b (375 mg) *n* = 41** |
| HVs with ≥ 1 TEAE* | 7 (63.6); 20 | 8 (88.9); 25 | 9 (90.0); 30 | 9 (100); 32 | 14 (93.3); 58 | 9 (100); 30 | 9 (100); 43 | 9 (100); 119 | 7 (77.8); 47 | 8 (88.9); 25 | 4 (40.0); 4 | 12 (30.8); 15 | 22 (53.7); 45 |
| Grade 1 or 2 severity† | 7 (63.6); 20 | 8 (88.9); 25 | 9 (90.0); 30 | 9 (100); 32 | 14 (93.3); 58 | 9 (100); 30 | 9 (100); 43 | 9 (100); 119 | 7 (77.8); 47 | 8 (88.9); 25 | 4 (40.0); 4 | 12 (30.8); 15 | 18 (43.9); 41 |
| Grade 3 or higher severity† | 0 | 0 | 1 (10.0); 1# | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HVs with ≥1 serious TEAE‡ | 0 | 0 | 1 (10.0); 1# | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HVs with ≥1 TEAE leading to trial discontinuation | 0 | 0 | 0 | 0 | 1 (6.7); 1** | 1 (11.1); 1** | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HVs with ≥1 treatment-related TEAE§ | 4 (36.4); 4 | 7 (77.8); 11 | 7 (70.0); 17 | 6 (66.7); 17 | 11 (73.3); 42 | 7 (77.8); 12 | 6 (66.7); 22 | 9 (100); 96 | 6 (66.7); 29 | 2 (22.2); 7 | 0 | 10 (25.6); 13 | 20 (48.8); 34 |
| *Most common TEAEs¶* | | | | | | | | | | | | | |
| Somnolence | 3 (27.3); 3 | 8 (88.9); 14 | 8 (80.0); 12 | 7 (77.8); 13 | 10 (66.7); 39 | 8 (88.9); 13 | 7 (77.8); 24 | 9 (100); 57 | 6 (66.7); 19 | 1 (11.1); 1 | 0 | 5 (12.8); 5 | 12 (29.3); 12 |
| Balance disorder | 0 | 0 | 4 (40.0); 4 | 0 | 0 | 0 | 0 | 7 (77.8); 27 | 2 (22.2); 2 | 0 | 0 | 0 | 1 (2.4); 1 |
| Cognitive disorder | 0 | 0 | 1 (10.0); 1 | 0 | 0 | 0 | 1 (11.1); 1 | 4 (44.4); 5 | 2 (22.2); 5 | 0 | 0 | 1 (2.6); 1 | 3 (7.3); 3 |
| Diarrhea | 0 | 0 | 0 | 0 | 1 (6.7); 1 | 1 (11.1); 1 | 0 | 0 | 2 (22.2); 3 | 0 | 0 | 0 | 0 |
| Dizziness | 0 | 4 (44.4); 4 | 0 | 1 (11.1); 1 | 2 (13.3); 3 | 0 | 0 | 1 (11.1); 1 | 2 (22.2); 2 | 0 | 0 | 1 (2.6); 1 | 8 (19.5); 8 |
| Euphoric mood | 0 | 0 | 1 (10.0); 1 | 0 | 0 | 0 | 0 | 2 (22.2); 2 | 1 (11.1); 1 | 0 | 0 | 0 | 0 |
| Fatigue | 1 (9.1); 1 | 0 | 0 | 0 | 1 (6.7); 1 | 0 | 0 | 0 | 0 | 2 (22.2); 2 | 0 | 1 (2.6); 1 | 2 (4.9); 2 |
| Headache | 4 (36.4); 5 | 1 (11.1); 1 | 1 (10.0); 1 | 2 (22.2); 2 | 0 | 1 (11.1); 1 | 1 (11.1); 1 | 2 (22.2); 2 | 2 (22.2); 3 | 0 | 0 | 4 (10.3); 4 | 5 (12.2); 6 |
| Insomnia | 0 | 0 | 0 | 0 | 2 (13.3); 2 | 0 | 0 | 0 | 0 | 2 (22.2); 2 | 0 | 0 | 0 |
| Lethargy | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 (22.2); 2 | 0 | 0 | 0 | 0 | 1 (2.4); 1 |
| Migraine | 0 | 0 | 0 | 2 (22.2); 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Vomiting | 0 | 0 | 0 | 0 | 0 | 1 (11.1); 1 | 0 | 0 | 0 | 0 | 0 | 0 | 2 (4.9); 2 |
| Dermatitis contact | 0 | 0 | 1 (10.0); 1 | 0 | 3 (20.0); 3 | 2 (22.2); 3 | 2 (22.2); 3 | 0 | 1 (11.1); 1 | 0 | 0 | 0 | 2 (4.9); 2 |

(*Continued*)





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(Continued)

| n (%); m | Part 1: SAD | | | Part 2: FE | | Part 3: MAD | | | | | Part 4: TSST | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cohort 1 (PBO, 70, 140, and 280 mg) $n = 11$ | Cohort 2 (PBO, 280, 420, and 560 mg) $n = 9$ | Cohort 2b (PBO, 750 and 1000 mg) $n = 10$ | Cohort 3 (140 mg) $n = 9$ | Pooled PBO $n = 15$ | Cohort 4 (250 mg QAM) $n = 9$ | Cohort 5 (375 mg QAM) $n = 9$ | Cohort 6 (500 mg QAM) $n = 9$ | Cohort 7 (375 mg QHS $n = 9$ | Cohort 8 (375 mg QHS) $n = 9$ | Cohort 9a (PBO) $n = 10$ | Cohort 9b (PBO) $n = 39$ | Cohort 9b (375 mg) $n = 41$ |
| Vascular access site bruising | 0 | 1 (11.1); 1 | 2 (20.0); 3 | 1 (11.1); 1 | 1 (6.7); 2 | 4 (44.4); 4 | 1 (11.1); 1 | 1 (11.1); 1 | 0 | 2 (22.2); 5 | 0 | 0 | 0 |
| Vascular access site pain | 2 (18.2); 2 | 0 | 1 (10.0); 1 | 3 (33.3); 3 | 2 (13.3); 2 | 2 (22.2); 2 | 3 (33.3); 3 | 2 (22.2); 3 | 2 (22.2); 4 | 2 (22.2); 3 | 0 | 0 | 0 |
| Vessel puncture site bruise | 0 | 2 (22.2); 3 | 0 | 1 (11.1); 1 | 0 | 0 | 1 (11.1); 1 | 2 (22.2); 4 | 1 (11.1); 1 | 1 (11.1); 1 | 0 | 0 | 1 (2.4); 1 |

*TEAEs were defined as adverse events that started on or after the first dose of study medication until the final safety follow-up at 14 ± 2 days after the last study drug administration. If a healthy volunteer had multiple occurrences of a TEAE, the healthy volunteer is presented only once in the count (*n*) column for a given system organ class and preferred term. TEAEs were coded according to the Medical Dictionary for Regulatory Activities version 24.1. †TEAE severity was graded by the investigator and defined as mild (grade 1), moderate (grade 2), severe (grade 3), life-threatening (grade 4), or death (grade 5). ‡Defined as a TEAE that fulfilled >1 of the following conditions: death; was immediately life-threatening; required inpatient or prolongation of existing hospitalization; resulted in persistent or significant disability or incapacity; resulted in a congenital anomaly or birth defect; or was a medically important event that jeopardized the healthy volunteer or required medical intervention to prevent an outcome listed above. §Determined by the investigator as "possibly," "probably," or "definitely" related to study treatment. ¶Occurred in ≥2 healthy volunteers in any cohort. #Healthy volunteer (750 mg of GlyphAllo) had a hand (digit) fracture during the washout week after period 1 that required surgical intervention; the event was determined by the investigator to be grade 3 and unrelated to treatment. **One healthy volunteer each in the pooled placebo group and cohort 4 (250 mg of GlyphAllo) discontinued the trial because of a TEAE of COVID-19 infection on days 5 and 3, respectively; each event was determined by the investigator to be grade 2 and unrelated to treatment.

the investigator to be unrelated to treatment, was reported [cohort 2b, $n = 1$ (750 mg of GlyphAllo)] and occurred during the washout week after period 1. All other TEAEs were mild or moderate in severity, and no other serious TEAEs were reported. No TEAEs resulted in trial discontinuation except for the two healthy volunteers mentioned above who discontinued the trial because of unrelated TEAEs of COVID-19 infection. Across the part 1 single-ascending dose cohorts, dose-dependent increases were observed in the incidence and number of TEAEs. In part 2, the incidence and number of TEAEs were similar across the high fat (HF)–fed, low fat (LF)–fed, and fasted states. Across the multiple-ascending dose cohorts in part 3, proportions of healthy volunteers with ≥1 TEAE were similar, but TEAE frequency increased with higher GlyphAllo doses. In part 4, reported TEAEs were consistent with those reported for cohorts 1 to 8, with more TEAEs reported for the GlyphAllo group versus the placebo group in cohort 9b.

The most common TEAE across all GlyphAllo doses and placebo was somnolence ($n = 75$; Table 2), which was mild or moderate and transient in all cases, increased in frequency with dose, and typically occurred in the first several hours after dosing. No TEAEs of loss of consciousness, excessive sedation, syncope, or hypoxia were observed, and no treatment-related severe or serious TEAEs, including clinically important hepatic, cardiac, or renal TEAEs, were reported. Central nervous system (CNS) events and psychiatric events were observed at higher GlyphAllo doses, but no dose-limiting toxicity was observed. No other clinically relevant findings were observed relating to clinical laboratory parameters, vital signs, electrocardiograms, or physical examinations for any treatment arm.

### PK outcomes

After single-ascending oral GlyphAllo doses (70 to 1000 mg) in healthy volunteers in part 1, mean plasma ALLO concentrations increased with dose (Fig. 4A), and maximum plasma concentrations ($C_{max}$) were reached between 4 and 5 hours postdose. Median time to $C_{max}$ ($T_{max}$) and mean apparent terminal half-life ($t_{1/2}$) for ALLO were not affected by dose and, although exhibiting some variability between doses, did not display any obvious trends (table S2). For ALLO, area under the plasma concentration–time curve (AUC) from time zero (time of dosing) extrapolated to infinity ($AUC_{0\text{-inf}}$), and $C_{max}$ increased with increasing dose within the clinical dosing range (<750 mg of GlyphAllo). These findings remained consistent after correction for dose concentration.

Part 2 of the clinical trial was not powered to formally assess food effect in terms of bioequivalence, but a mild food effect on plasma ALLO concentrations was observed after administration of a single oral dose of GlyphAllo (140 mg) under both HF-fed and LF-fed conditions (Fig. 4B and table S3). The LF-fed state exhibited a higher apparent effect versus the HF-fed state, but this may have been related to the small sample size. These changes may not be clinically relevant given the overall magnitude of the observed effect.

After multiple-ascending oral doses of GlyphAllo (250 to 500 mg) in healthy volunteers in part 3, ALLO (Fig. 4, C and D, and table S4) exhibited minimal accumulation and reached a steady state by days 2 and 3 of once-daily administration. Mean plasma ALLO concentrations increased with increasing GlyphAllo dose, and mean plasma $C_{max}$ values were higher on day 1 than on day 7 across all cohorts. The median $T_{max}$ was ~3 to 4 hours postdose and altered little after repeat dosing. No apparent differences were observed between once daily in the morning and once daily in the evening dosing. Accumulation ratios suggested no notable accumulation of ALLO in

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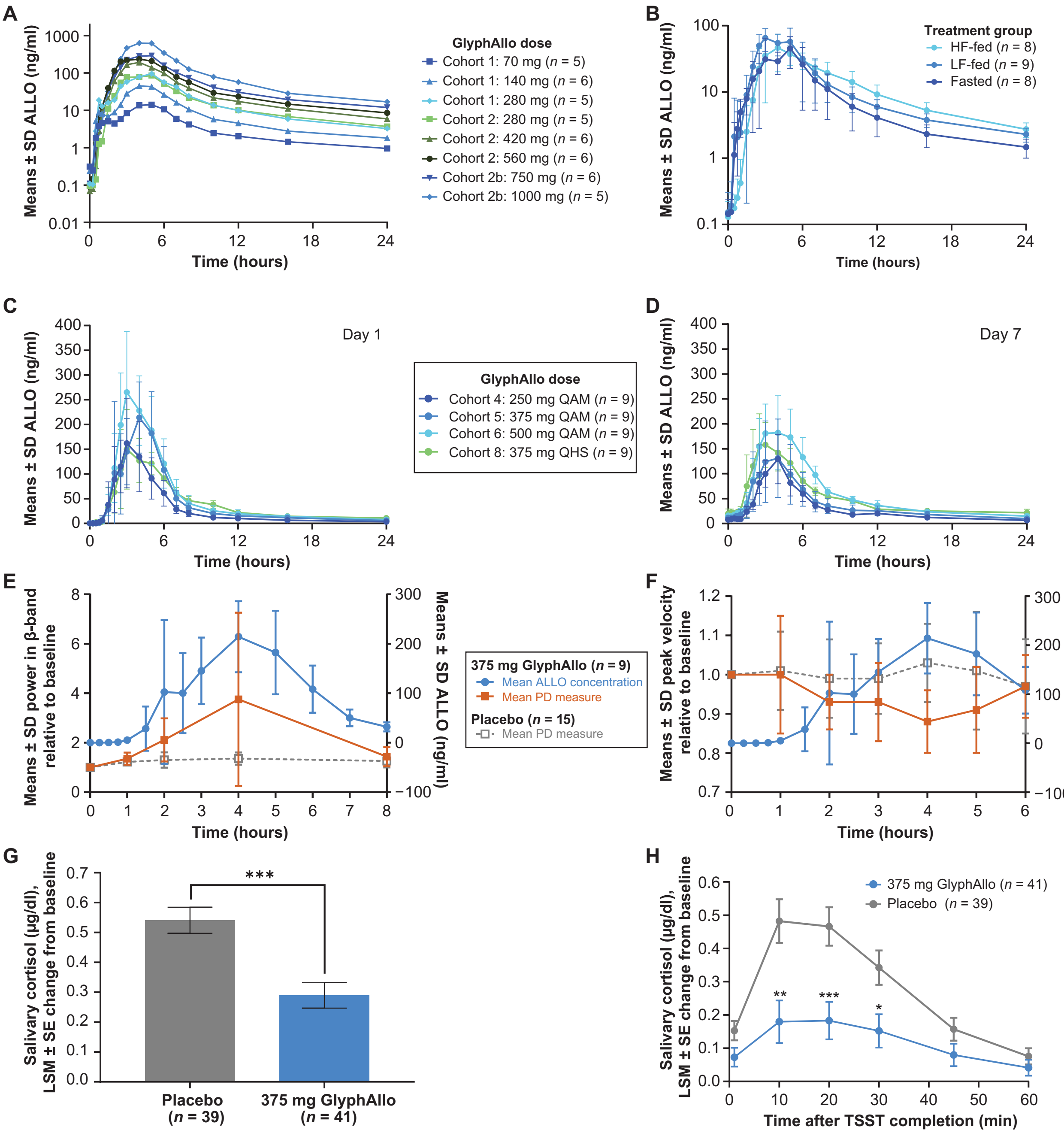


**Fig. 4. Pharmacokinetic and pharmacodynamic outcomes in a phase 1/2a clinical trial in healthy volunteers.** (**A**) Semilogarithmic concentration-time profile of ALLO (means ± SD) after oral GlyphAllo administration in single-ascending dose cohorts on day 1. (**B**) Semilogarithmic concentration-time profile of ALLO (means ± SD) after single oral administration of 140 mg of GlyphAllo in fasted and fed states. Linear concentration-time profile of ALLO (means ± SD) after oral GlyphAllo administration in multiple-ascending dose cohorts on (**C**) day 1 and (**D**) day 7. (**E**) Power (means ± SD) in qEEG β-band (eyes open condition) relative to baseline (left axis, orange line and gray dashed line) over time versus ALLO concentration-time profile (right axis, blue line) after oral administration of placebo or 375 mg of GlyphAllo in multiple-ascending dose cohorts on day 1. (**F**) Saccade peak velocity (means ± SD) relative to baseline (left axis, orange line and gray dashed line) over time versus ALLO concentration-time profile (right axis, blue line) after oral administration of placebo or 375 mg of GlyphAllo in multiple-ascending dose cohorts on day 1. (**G**) Peak salivary cortisol concentrations [least squares mean (LSM) ± SE; $\log_{10}$ transformed] and (**H**) salivary cortisol concentrations (LSM ± SE) over time relative to baseline following TSST completion after oral administration of a single dose of placebo or 375 mg of GlyphAllo. *$P < 0.05$; **$P \leq 0.01$; ***$P \leq 0.001$ versus placebo. Analysis of variance (ANOVA) was used in (G), and MMRM was used in (H). QAM, once daily in the morning; QHS, once daily in the evening.

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any dose cohort, and no indication of meaningful loss or gain in exposure was observed upon repeated dosing for 7 days. For all cohorts, GlyphAllo reached a steady state by days 2 or 3 and was independent of dose administered and timing.

In parts 1 to 3 of the clinical trial, PK time courses for the ALLO combined ($ALLO_{combined}$) analyte (tables S5 and S6) were generally similar to those for ALLO. $C_{max}$ values for $ALLO_{combined}$ were ≈1 order of magnitude greater than peak values for ALLO, reflecting that the TG-bound ALLO prodrug was contained in chylomicrons, which are restricted to plasma circulation and thus have a far lower volume of distribution than that of ALLO (*36*). Overall, PK trends were consistent between the two analytes. In part 4, a single sample was collected from each healthy volunteer in cohort 9b at ~3 to 4 hours postdose, indicating a median (range) plasma ALLO concentration of 144 (38.1, 451) ng/ml; this concentration was consistent with that observed for the equivalent 375-mg dose (cohort 5) at 4 hours in part 3 of the clinical trial [median (range), 144 (4.43, 264) ng/ml].

We conducted allometric scaling on PK parameters on the basis of three species (rat, NHP, and dog; table S7). Preliminary analysis suggested that NHP data predicted human PK parameters well compared with other species. Allometric scaling using all three species suggested that apparent clearance was overpredicted in humans, whereas the apparent volume of distribution of the terminal phase was underpredicted. Further interspecies scaling and PK modeling work are required to properly characterize these interspecies differences.

### *PD assessments*

To investigate the association between PK and PD effects of GlyphAllo at doses of 250, 375, and 500 mg, PD markers of $GABA_A$ receptor engagement were assessed in part 3, including quantitative electroencephalogram (qEEG) spectral changes and saccadic eye velocity (SEV) assessed by video-oculography. Compared with placebo, GlyphAllo dosing increased qEEG beta power relative to baseline, reaching a maximum at 4 hours postdose and mirroring increasing plasma ALLO concentrations (Fig. 4E); these findings were more notable at the highest doses tested (fig. S7). Similarly, SEV showed dose-dependent changes with GlyphAllo administration, and ratios to baseline for peak SEV decreased with increasing plasma ALLO concentrations (Fig. 4F), with maximal SEV reductions at 4 hours postdose that were more prominent with higher GlyphAllo doses (fig. S7C).

To further probe PD activity beyond the markers described above, we conducted a randomized, placebo-controlled trial (part 4) of the effect of a single 375-mg dose of GlyphAllo on the TSST, a clinically validated test for anxiety in healthy volunteers involving exposure to unpredictable and anticipatory social stress and subsequent measurements of cortisol, a key stress hormone (*50*, *51*). In the placebo group, TSST substantially increased salivary cortisol concentrations (Fig. 4G), peaking at 10 to 20 min post-TSST (Fig. 4H) and consistent with the known delay in cortisol response to a stressor (*51*). In contrast, GlyphAllo treatment 3.5 hours before the stressor potently blunted the peak salivary cortisol response [log (base 10)–transformed maximal change from baseline] compared with placebo ($P = 0.0001$) (Fig. 4G). The maximum nontransformed mean salivary cortisol concentration was significantly decreased by 2.2-fold with GlyphAllo treatment versus placebo ($P = 0.0016$), with between-group differences ranging from a 62.8% reduction at 10 min post-TSST to 45.5% reduction at 60 min post-TSST with GlyphAllo versus placebo. Thus, GlyphAllo-treated healthy volunteers exhibited a markedly blunted stress response at all time points as measured by salivary cortisol compared with placebo. Self-reported assessments of anxiety were also collected using the State-Trait Anxiety Inventory, which confirmed that the TSST transiently and modestly increased self-reported anxiety in both the placebo and GlyphAllo groups (fig. S8). These results are consistent with previous data showing that healthy volunteers are less likely to self-report anxiety symptoms after the TSST compared with patients with an anxiety disorder, who may experience a heightened negative psychological response to social stress (*52*). These results are also consistent with previously published self-reported assessments during the TSST using known anxiolytic drugs such as alprazolam in healthy volunteers, in whom self-reported anxiety increased in both drug and placebo groups in response to the TSST (*50*) (see Supplementary Materials and Methods for full methods).

## DISCUSSION

ALLO shows considerable promise as a therapeutic agent but has not been approved in an oral form owing to low bioavailability due to rapid first-pass hepatic metabolism (*28*, *29*, *39*, *40*, *53*). Orally bioavailable synthetic ALLO analogs have been advanced into clinical trials (*54*, *55*) but are chemically distinct and thus may exhibit different pharmacological properties than endogenous ALLO (*28*, *55*). The lymphatically targeted Glyph platform described here uses a TG-mimetic prodrug approach in combination with appropriate SI linkers to boost lymphatic transport (>40% dose), avoid first-pass metabolism, and enable systemic drug exposure.

Previous reports of this approach have shown a proof of concept in rodents and larger animals and indicate utility of the Glyph platform across other active pharmaceutical ingredients and targets (*18*, *19*, *21*–*23*, *25*). Other methods to enhance lymphatic transport via oral administration typically include coadministration with lipid formulations and the use of simpler prodrug approaches such as lipophilic esters and directly attached glyceride and phospholipid prodrugs (*1*, *4*). However, these had moderate success, and the extent of lymphatic transport of the only lymph-directed marketed prodrug (testosterone undecanoate) is <5% of the administered dose (*19*, *56*, *57*).

A variety of ALLO prodrugs were generated with varying linkers and SI groups. Linker substituents were used to modify ester bond stability at both ends of the linker, and the presence of a methyl group in the β-position to the ester bond added stability and slowed esterase-mediated bond cleavage (*22*, *23*). Here, a five-carbon linker with a 3-methyl substituent imparted protection at both ends of the linker, enhancing gut stability (Fig. 1D), whereas prodrugs with an unsubstituted C10 linker were much less GI stable. The SI group is critical for promoting active pharmaceutical ingredient plasma release once the prodrug has been digested, absorbed, reesterified, and transported to the systemic circulation via the lymph (*19*). For all ALLO prodrugs evaluated here, lymphatic transport of 40 to 60% was observed (Fig. 1E), demonstrating >50-fold improvements over unmodified ALLO, consistent with previous applications of this platform (*18*, *19*, *21*). Consistently good lymphatic transport and plasma release was associated with robust oral exposure. This was particularly the case in dogs and NHPs, in which only the SI-containing prodrugs **5** and **6** showed good in vitro ALLO release (Fig. 2, G and K) and accompanying oral exposure (Fig. 2, H and L). Prodrug **6** was subsequently chosen as the clinical candidate (GlyphAllo).

In a first-in-human phase 1/2a clinical trial, oral dosing of GlyphAllo resulted in therapeutically relevant ALLO exposures that were nearly

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an order of magnitude greater (ninefold) than previously published values of oral ALLO dosing (*36*), validating the ability of the Glyph platform to enhance oral bioavailability. Combined with preclinical evidence, these findings suggest that oral GlyphAllo redirects drug transport pathways away from the portal blood and to the intestinal lymph, avoiding first-pass hepatic metabolism and delivering reesterified prodrug into the systemic circulation, where prodrug conversion leads to quantifiable ALLO concentrations in plasma.

GlyphAllo displayed dose proportional plasma exposure that diminished only at higher doses of 750 to 1000 mg. Furthermore, the absorption kinetics of GlyphAllo mirrored previously published reports of the kinetics of chylomicron appearance in plasma (i.e., showing a 1- to 2-hour lag followed by a 3- to 5-hour $T_{\max}$) (*58*). We observed a minimal and positive effect of food on the observed PK profiles of GlyphAllo, and the LF-fed state had a larger apparent effect than the HF-fed state. Typically, lymphatic transport is enhanced after dosing with food because this supplies a large quantity of fat to drive chylomicron formation (*59*). We have previously shown that coadministration with a lipid formulation can enhance lymph transport above that of fasted conditions, but this usually does not reach transport levels seen under fed conditions (*60*). It is possible that the minimal food effects seen here reflect lipid formulation effects or high efficiency of chylomicron incorporation of the TG prodrug, but further mechanistic studies would be required to confirm this effect.

GlyphAllo was generally well tolerated. A dose-dependent increase in the number of TEAEs was observed, but no dose-limiting toxicity was reached. Drug-related TEAEs were overall consistent with the pharmacological profile of ALLO (clinical trial parts 1 to 4) and with the situational stressor of the TSST itself (clinical trial part 4) (*34*, *35*, *50*, *51*). Across the four parts of the clinical trial, the most common TEAE was mild, transient somnolence, consistent with the pharmacology of ALLO. Single-ascending dose and multiple-ascending dose GlyphAllo dosing resulted in minimal food effect and no notable accumulation from daily dosing. The estimated $T_{\max}$ was between 3 and 4 hours, with a terminal half-life after single or multiple doses of ≈24 hours. On the basis of these observed PK and TEAE findings, GlyphAllo is likely suitable for evening dosing.

The assessed PD markers of $GABA_A$ receptor engagement, beta EEG power and SEV, have been previously shown to be sensitive to GABA receptor potentiation (*61*, *62*). After GlyphAllo administration, dose-dependent changes in these PD biomarkers were correlated with plasma ALLO concentrations, and maximum effects were observed at 3 to 4 hours after administration, corresponding to the $C_{\max}$. Thus, GlyphAllo oral dosing resulted in therapeutically relevant systemic and brain ALLO concentrations and provided $GABA_A$ receptor engagement in a dose- and exposure-dependent manner, with no meaningful hysteresis between concentration and PD effects. This suggests that ALLO after oral GlyphAllo administration may have similar target engagement in the CNS as intravenous infusion of ALLO.

We also assessed the effect of a single GlyphAllo dose in the TSST, a clinically validated model of anxiety commonly used to evaluate the effects of pharmacotherapeutics such as benzodiazepines and antidepressants on the stress response (*50*, *51*, *63*). The TSST was designed to simulate situational characteristics of novelty, uncertainty, and social-evaluative threat, activating the hypothalamic-pituitary-adrenal (HPA) axis and increasing concentrations of the stress hormone cortisol, blood pressure, and other markers. Endogenous ALLO concentrations have previously been shown to increase after the TSST (*51*), suggesting that ALLO acts as an endogenous stress regulatory hormone. In this trial, the TSST substantially increased salivary cortisol among healthy volunteers who received a placebo, peaking at 10 to 20 min post-TSST, consistent with the known delayed cortisol response to a stressor (*51*). GlyphAllo substantially reduced salivary cortisol at all post-TSST time points, meeting the primary end point for that portion of the trial and demonstrating that GlyphAllo regulates HPA axis reactivity to acute stress in humans, as has been reported for ALLO in rodent models (*64*). The effects of GlyphAllo on salivary cortisol concentrations were comparable to the potent benzodiazepine anxiolytic alprazolam (*50*, *51*), although, unlike alprazolam, the $GABA_A$ receptor modulatory pharmacology of GlyphAllo stems from that of ALLO. Thus, these findings confirm the pharmacological activity of GlyphAllo and support its further investigation in stress-related mood and anxiety disorders, including major depressive disorder with anxious distress.

Our study has some limitations. Our findings demonstrate the ability of GlyphAllo to bypass liver metabolism in multiple species. However, gut lymphatic transport was not measured directly in higher-order species and humans. Inherent to an early phase trial in healthy volunteers, interpretation of findings such as the minimal food effect were limited by trial population and resulting variability, which may affect the generalizability and clinical relevance of our findings. The evidence we present here is consistent with a major role for gut lymphatic transport of the prodrug in driving the systemic circulation of ALLO, but we cannot rule out some minimal absorption of ALLO via the blood. Furthermore, we did not conduct a head-to-head trial comparing GlyphAllo with an intravenous formulation of ALLO. However, comparisons to previously published clinical studies of non-Glyph oral formulations of ALLO support our finding of therapeutically relevant systemic ALLO concentrations after oral GlyphAllo dosing in healthy volunteers due to enhanced bioavailability (*36*).

In summary, we synthesized and evaluated a series of lipid prodrugs to redirect drug transport through the gut lymphatic system and bypass first-pass hepatic metabolism, thereby allowing effective oral delivery of the neuroactive steroid ALLO to the systemic circulation. After differentiation of prodrug performance using in vitro and in vivo assays in small- and large-animal models, an orally bioavailable ALLO prodrug, GlyphAllo, was advanced into clinical development in a multipart phase 1/2a clinical trial. In healthy adult volunteers, GlyphAllo was well tolerated and demonstrated generally dose-proportional PK and PD. The translation and validation of the Glyph prodrug platform open opportunities to apply the technology to other small-molecule drugs that could benefit from enhanced lymphatic absorption, including those with limited oral bioavailability due to extensive first-pass metabolism or other limitations (*4*).

## MATERIALS AND METHODS

### Clinical trial design

Prodrug synthesis was carried out from known 1,3-dioleoylglycerol (*65*) using previously described methods (fig. S2 and table S1) (*7*, *8*). The following experiments were conducted using previously described methods (*7*): in vitro prodrug hydrolysis by pancreatin digestion in simulated intestinal fluid; lymphatic transport assay in adult male Sprague-Dawley rats; in vitro release of active pharmaceutical ingredient and intermediates from prodrugs in rats, adult

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male beagle dogs, NHPs (adult male cynomolgus macaques), and human plasma supplemented with lipoprotein lipase; and PK studies in rats, dogs, and NHPs. For all experiments, prodrug and prodrug-related compounds were quantified using liquid chromatography–mass spectrometry. Rat studies were unblinded and nonrandomized. Animals were only excluded due to surgical failure. Dog and NHP studies were unblinded. Data from two dogs were excluded from analysis due to poor ALLO metabolism compared to intravenously dosed ALLO. No other outliers were excluded in the dog or NHP studies.

Following selection of the prodrug **6** for clinical testing, a multipart, phase 1/2a, first-in-human clinical trial (NCT05129865) was initiated to assess safety, tolerability, and PK of oral GlyphAllo in 189 healthy volunteers (fig. S6, A to D). Part 1 was a randomized, double-blind, placebo-controlled single-ascending dose cohort using a three-period, three-sequence, crossover dose-escalation design (Table 3). Single administration of GlyphAllo or placebo was administered orally in the morning. Cohort 1 was dosed following an overnight fast of ≥10 hours, and cohorts 2 and 2b were dosed in a fed state after a standard meal. Study drug administration was separated by a washout period of ≥7 days.

Part 2 was a randomized, open-label, three-period, three-sequence, crossover cohort study in healthy volunteers to assess the effects of food on the PK, safety, and tolerability of oral GlyphAllo (140 mg; Table 3). In a single cohort (cohort 3), dosing was arranged in one of three sequences such that each participant would receive three GlyphAllo administrations (randomized to fasted, LF fed, or HF fed) separated by a washout period of ≥7 days.

Part 3 was a randomized, double-blind, placebo-controlled, sequential multiple-ascending dose cohort study in healthy volunteers to assess the safety, tolerability, and PK of multiple doses (up to 7 days) of oral GlyphAllo (Table 3). Trial participants in cohorts 4, 5, and 6 received the study drug once daily in the morning in a fed state, and participants in cohorts 7 and 8 received the study drug once daily in the evening in a fed state.

Part 4 was a randomized, double-blind, placebo-controlled trial in healthy volunteers to evaluate the efficacy of oral GlyphAllo in the TSST, a clinically validated test of anxiety in response to stress (Table 3). For cohort 9a (validation cohort), 10 single-blind participants received placebo and underwent all part 4 procedures for the purpose of validating the TSST. For cohort 9b (randomized cohort), participants were randomized in a 1:1 ratio to receive a single dose of placebo or 375 mg of GlyphAllo in the morning following an overnight fast of ≥10 hours.

The first healthy volunteer was enrolled on 30 November 2021, and the last healthy volunteer completed on 19 December 2022. All healthy volunteers were enrolled at a single center in Australia (CMAX Level 5, Adelaide, SA, Australia). The protocol, protocol amendments, and other relevant documents were reviewed and approved by the local Institutional Review Board/Human Research Ethics Committee, the Bellberry Human Research Ethics Committee (EC00459;

**Table 3. Cohorts, treatment periods, and dosing schema for phase 1/2a trial in healthy volunteers.**

| Study part | Cohort | Number of HVs (planned)[*] | | Treatment sequence | Treatment period and GlyphAllo dosing | | |
|---|---|---|---|---|---|---|---|
| | | GlyphAllo | Placebo | | Period 1 | Period 2 | Period 3 |
| Part 1: SAD[†] (crossover) | 1 | 3 | | 1 | Placebo | 140 mg | 280 mg |
| | | 3 | | 2 | 70 mg | Placebo | 280 mg |
| | | 3 | | 3 | 70 mg | 140 mg | Placebo |
| | 2 | 3 | | 1 | Placebo | 420 mg | 560 mg |
| | | 3 | | 2 | 280 mg | Placebo | 560 mg |
| | | 3 | | 3 | 280 mg | 420 mg | Placebo |
| | 2b | 3 | | 1 | Placebo | 1000 mg | 1250 mg[‡] |
| | | 3 | | 2 | 750 mg | Placebo | 1250 mg[‡] |
| | | 3 | | 3 | 750 mg | 1000 mg | Placebo |
| Part 2: food effect[§] (crossover) | 3 | 3 | NA | NA | 140 mg; HF fed | 140 mg; LF fed | 140 mg; fasted |
| | | 3 | | | 140 mg; LF fed | 140 mg; fasted | 140 mg; HF fed |
| | | 3 | | | 140 mg; fasted | 140 mg; HF fed | 140 mg; LF fed |
| Part 3: MAD[¶] | 4 | 9 | 3 | NA | 250 mg of QAM for 7 days | | |
| | 5 | 9 | 3 | | 375 mg of QAM for 7 days | | |
| | 6 | 9 | 3 | | 500 mg of QAM for 7 days | | |
| | 7 | 9 | 3 | | 375 mg of QHS for 7 days | | |
| | 8 | 9 | 3 | | 375 mg of QHS for 7 days | | |
| Part 4: TSST[#] | 9a | NA | 10 | NA | NA | | |
| | 9b | 40 | 40 | | 375 mg or placebo | | |

*A total of 122 distinct participants received ≥1 dose of GlyphAllo in the phase 1/2a trial across parts 1 to 4. However, a total of 139 participants appear in the safety data tables under GlyphAllo because of their inclusion in ≥1 cohort as per the crossover design. †The starting GlyphAllo dose for cohort 1 in period 1 was 70 mg. Washout between periods was ≥7 days. Cohort 1 was dosed in a fasted state, and cohorts 2 and 2b were dosed in a fed state after a standard meal. ‡Initial pharmacokinetic data for 1000 mg suggested that exposure was greater than dose-proportional compared with the 750-mg dose; period 3 dosing at 1250 mg did not proceed. §Dose level was selected on the basis of available data from completed part 1 (single-ascending dose). ¶Cohorts 4 to 8 were dosed in a fed state. #Treatment was administered orally by a single dose of either 375 mg of GlyphAllo or placebo based on cohort (cohort 9a) or randomization (cohort 9b) in the morning following an overnight fast of ≥10 hours.

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Eastwood, SA, Australia) before trial initiation. The trial was conducted in accordance with the Declaration of Helsinki, the National Health and Medical Research Council National Statement on Ethical Conduct in Human Research, the current Good Clinical Practice Guideline of the International Council for Harmonisation, and appropriate associated national and local legal and regulatory requirements. All trial participants provided written informed consent before initiation of any trial procedures.

### Clinical trial participants

Males or females aged 18 to 55 years (inclusive) were eligible for the clinical trial if identified as healthy based on medical history, physical examination, vital signs, 12-lead electrocardiograms, and clinical laboratory evaluations. Clinical trial participants were required to have a body mass index between 17.5 and 30.5 kg/m$^2$ (inclusive), a total body weight of >50 kg, and no clinically significant hematologic, renal, endocrine, pulmonary, GI, cardiovascular, hepatic, psychiatric, neurologic, or allergic disease. Full inclusion and exclusion criteria are listed in Supplementary Materials and Methods.

### Clinical assessments

Blood samples for PK analysis were collected at the time points delineated in Supplementary Materials and Methods. EEGs were collected at multiple time points before and after dosing on days 1 and 7 of the multiple-ascending dose portion of the trial for cohorts 4 to 7; for each session, 15-min recordings were collected while healthy volunteers were sitting quietly, to include resting brain activity with eyes open and eyes closed. Video-oculography was used to measure saccades, antisaccades, and pursuits; further details of video-oculography assessments can be found in Supplementary Materials and Methods. Safety assessments included TEAEs, physical examinations, clinical laboratory tests, vital sign measurements, and electrocardiograms.

### Statistical analysis

For parts 1 to 3, no formal calculation was made to determine sample size. For part 4, based on literature reports of changes in salivary cortisol concentrations resulting from performance of the TSST (*50*), the maximal treatment effect (Cohen's *d*) of GlyphAllo relative to placebo was estimated to be between 0.5 to 1.0. Assuming an effect size of 0.7, the total sample size of 80 participants ($n = 40$ in each of two treatment groups) was estimated to provide 80% power for a two-sided *t* test with an alpha level of 0.05.

Demographic and safety data were assessed in the safety population [defined as all participants who received any amount of the study drug (GlyphAllo or placebo)] and summarized descriptively. PK data were assessed in the PK population [defined as all participants who received any amount of active treatment (GlyphAllo) and had ≥1 quantifiable plasma concentration data point for the analyte being assessed]. PK parameters were determined from analyte plasma concentrations using noncompartmental methods via Phoenix WinNonlin (version 8.3; CertaraInc., Mountain View, CA, USA).

For part 1 (single-ascending dose), a linear regression model was constructed for inferential analysis of natural log-transformed PK parameters (see Supplementary Materials and Methods). For part 2 (food effect), PK parameters from the HF-fed, LF-fed, and fasted conditions were evaluated by a random effect analysis of covariance. For part 3 (multiple-ascending dose), qEEG parameters were analyzed using a mixed model for repeated measures (MMRM), and video-oculography data were summarized descriptively for actual and change from baseline values. For part 4, salivary cortisol changes resulting from the TSST have been reported to be nonnormally distributed (*50*, *66*). Therefore, the by-subject cortisol values were log (base 10) transformed to bring moderately skewed data to a more normally distributed dataset, achieving a more constant variance. Analysis of salivary cortisol concentrations was based on an analysis of covariance, with fixed effects for treatment and covariate pre-TSST cortisol values. Analyses were performed using the Statistical Analysis System (SAS Institute Inc., Cary, NC, USA) version 9.4.

## Supplementary Materials

**The PDF file includes:**

Materials and Methods
Figs. S1 to S8
Tables S1 to S7
Reference (*67*)

**Other Supplementary Material for this manuscript includes the following:**

Data file S1
MDAR Reproducibility Checklist

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# An oral allopregnanolone prodrug bypasses liver metabolism via lymphatic transport enabling bioavailability in animals and humans

Jamie S. Simpson, Tim Quach, Sifei Han, Luojuan Hu, Natalie L. Trevaskis, Nathania J. Leong, Garima Sharma, Dan Zheng, Steven M. Paul, Christopher J. H. Porter, Daniel K. Bonner, and Michael C. Chen

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**Editor's summary**
Many drugs cannot be given orally often because the liver breaks down the drug. In a new study, Simpson *et al.* developed the Glyph platform for lymphatic targeting of prodrugs in which a drug of interest is conjugated to a dietary lipid molecule, thus bypassing liver metabolism and shifting drug absorption to the gut lymphatic system. The authors use this platform to develop GlyphAllo, a prodrug of the neuroactive steroid antidepressant and anxiolytic drug allopregnanolone. They show through preclinical and clinical testing that oral GlyphAllo safely achieved therapeutic concentrations and reduced stress hormone responses, supporting its potential for treating mood and anxiety disorders. —Orla Smith

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